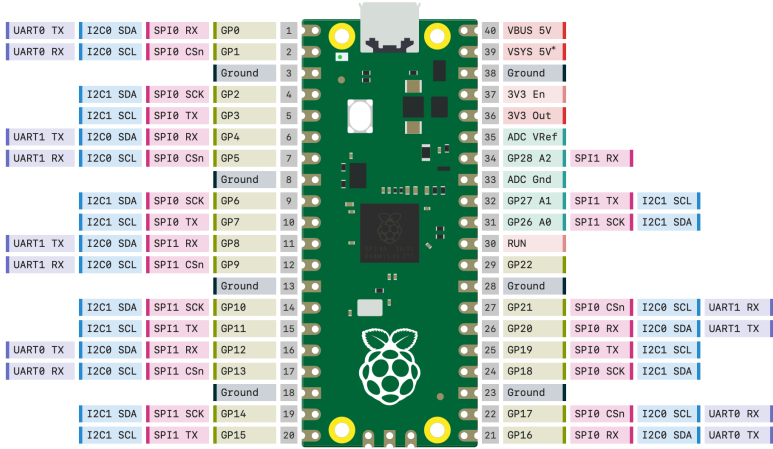
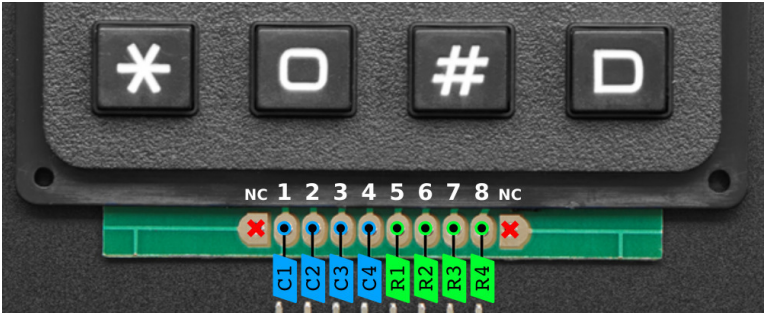
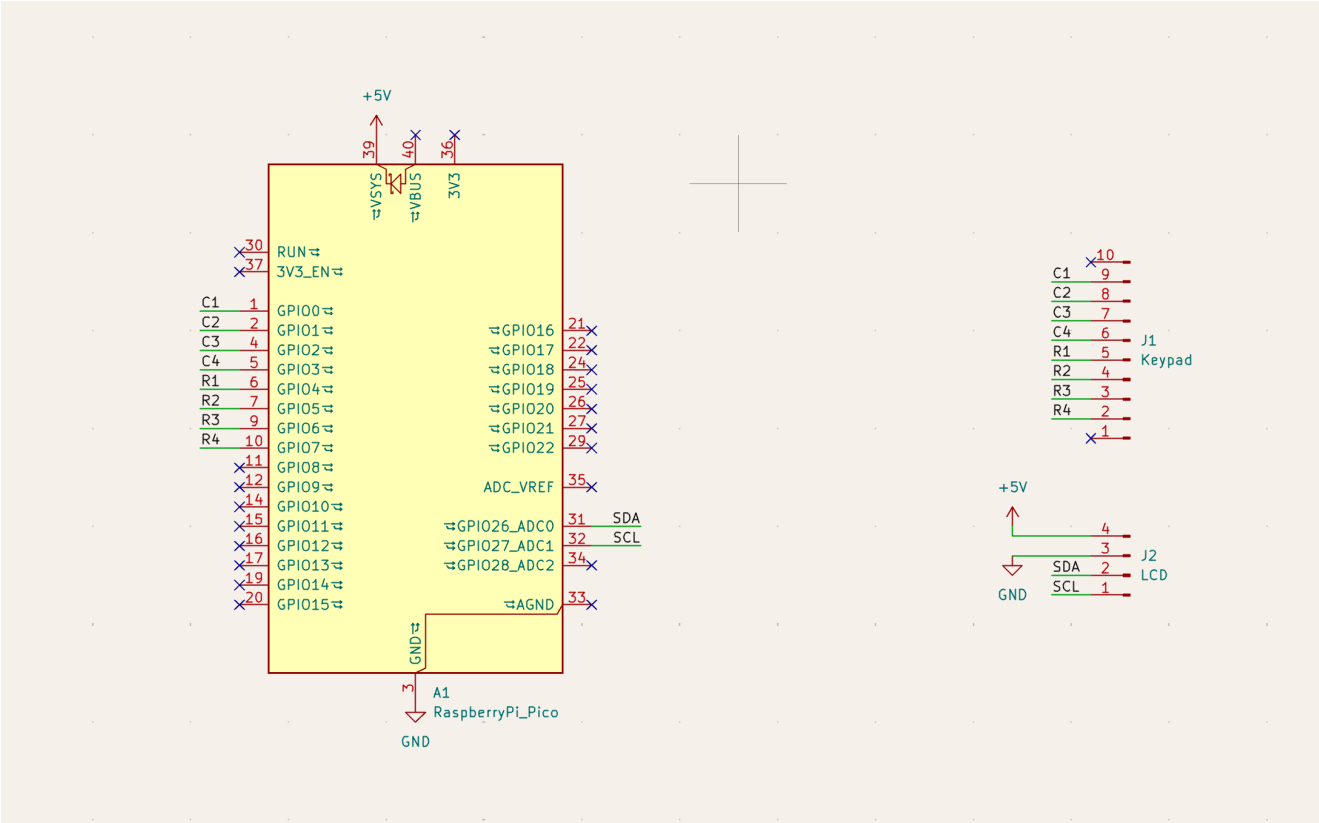


Connections



2.6.2. SRAM

There is a total of 264kB of on-chip SRAM. Physically this is partitioned into six banks, as this vastly improves memory bandwidth for multiple masters, but software may treat it as a single 264kB memory region. There are no restrictions on what is stored in each bank: processor code, data buffers, or a mixture. There are four 16k x 32-bit banks (64kB each) and two 1k x 32-bit banks (4kB each).

! IMPORTANT

Banking is a *physical* partitioning of SRAM which improves performance by allowing multiple simultaneous accesses. *Logically* there is a single 264kB contiguous memory.

Each SRAM bank is accessed via a dedicated AHB-Lite arbiter. This means different bus masters can access different SRAM banks in parallel, so up to four 32-bit SRAM accesses can take place every system clock cycle (one per master).

SRAM is mapped to system addresses starting at `0x20000000`. The first 256kB address region is word-striped across the four larger banks, which provides a significant memory parallelism benefits for most use cases.

Consecutive words in the system address space are routed to different RAM banks as shown in [Table 153](#).

Table 153. SRAM bank0/1/2/3 striped mapping.

System address	SRAM Bank	SRAM word address
<code>0x20000000</code>	Bank 0	0
<code>0x20000004</code>	Bank 1	0
<code>0x20000008</code>	Bank 2	0
<code>0x2000000c</code>	Bank 3	0
<code>0x20000010</code>	Bank 0	1
<code>0x20000014</code>	Bank 1	1
<code>0x20000018</code>	Bank 2	1
<code>0x2000001c</code>	Bank 3	1
<code>0x20000020</code>	Bank 0	2
<code>0x20000024</code>	Bank 1	2
<code>0x20000028</code>	Bank 2	2
<code>0x2000002c</code>	Bank 3	2
etc		

The next two 4kB regions (starting at `0x20040000` and `0x20041000`) are mapped directly to the smaller, 4kB memory banks. Software *may* choose to use these for per-core purposes, e.g. stack and frequently-executed code, guaranteeing that the processors never stall on these accesses. However, like all SRAM on RP2040, these banks have single-cycle access from *all* masters providing no other masters are accessing the bank in the same cycle, so it is reasonable to treat memory as a single 264kB device.

The four 64kB banks are also available at a non-striped mirror. The four 64kB regions starting at `0x21000000`, `0x21010000`, `0x21020000`, `0x21030000` are each mapped directly to one of the four 64kB SRAM banks. Software can explicitly allocate data and code across the physical memory banks, for improved memory performance in exceptionally demanding cases. This is often unnecessary, as memory striping usually provides sufficient parallelism with less software complexity.

The non-striped mirror starts at an offset of +16MB above the base of SRAM, as this is the maximum offset that allows ARMv6M subroutine calls between the smaller banks and the non-striped larger banks.